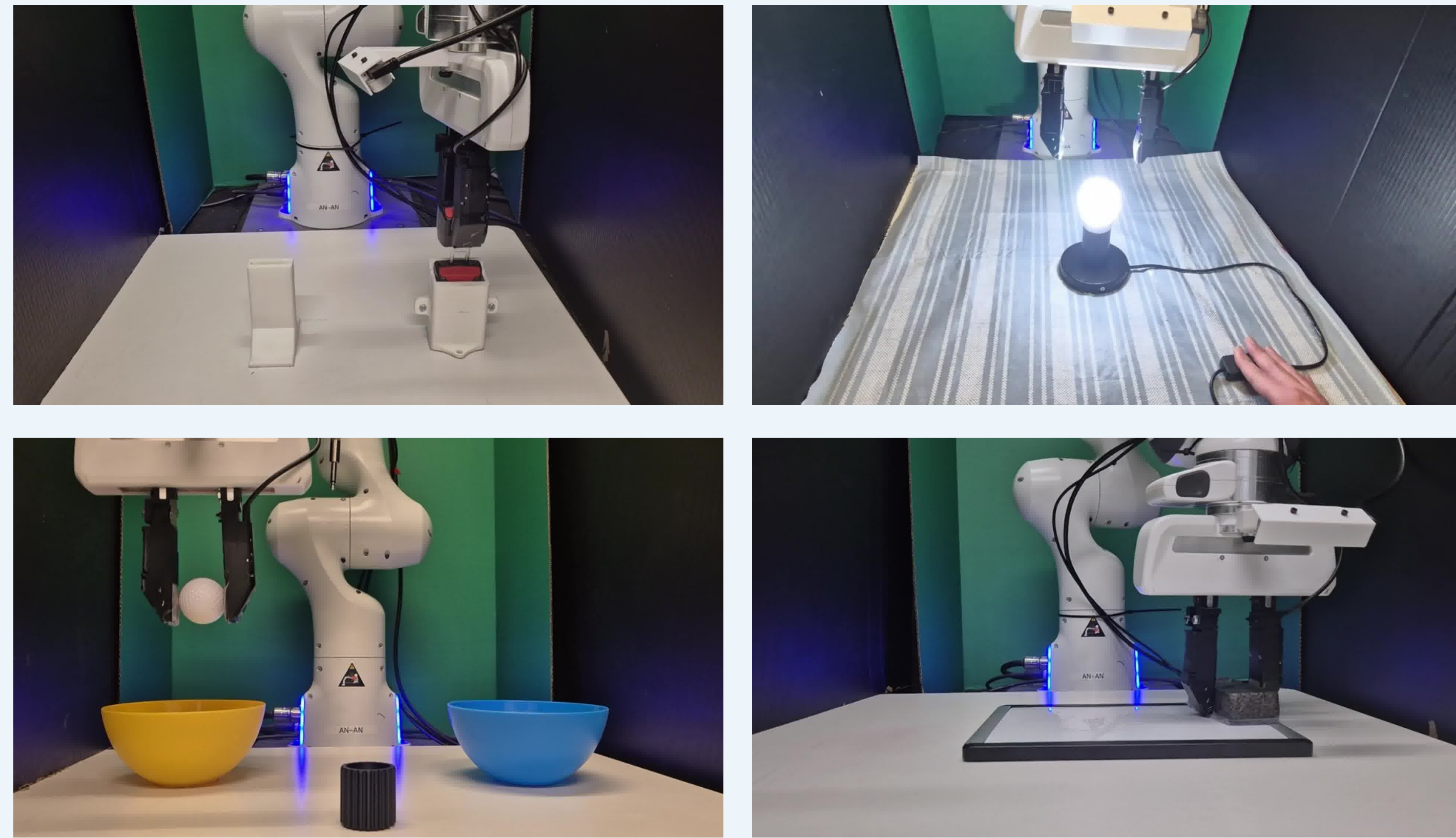


(a) Motivation

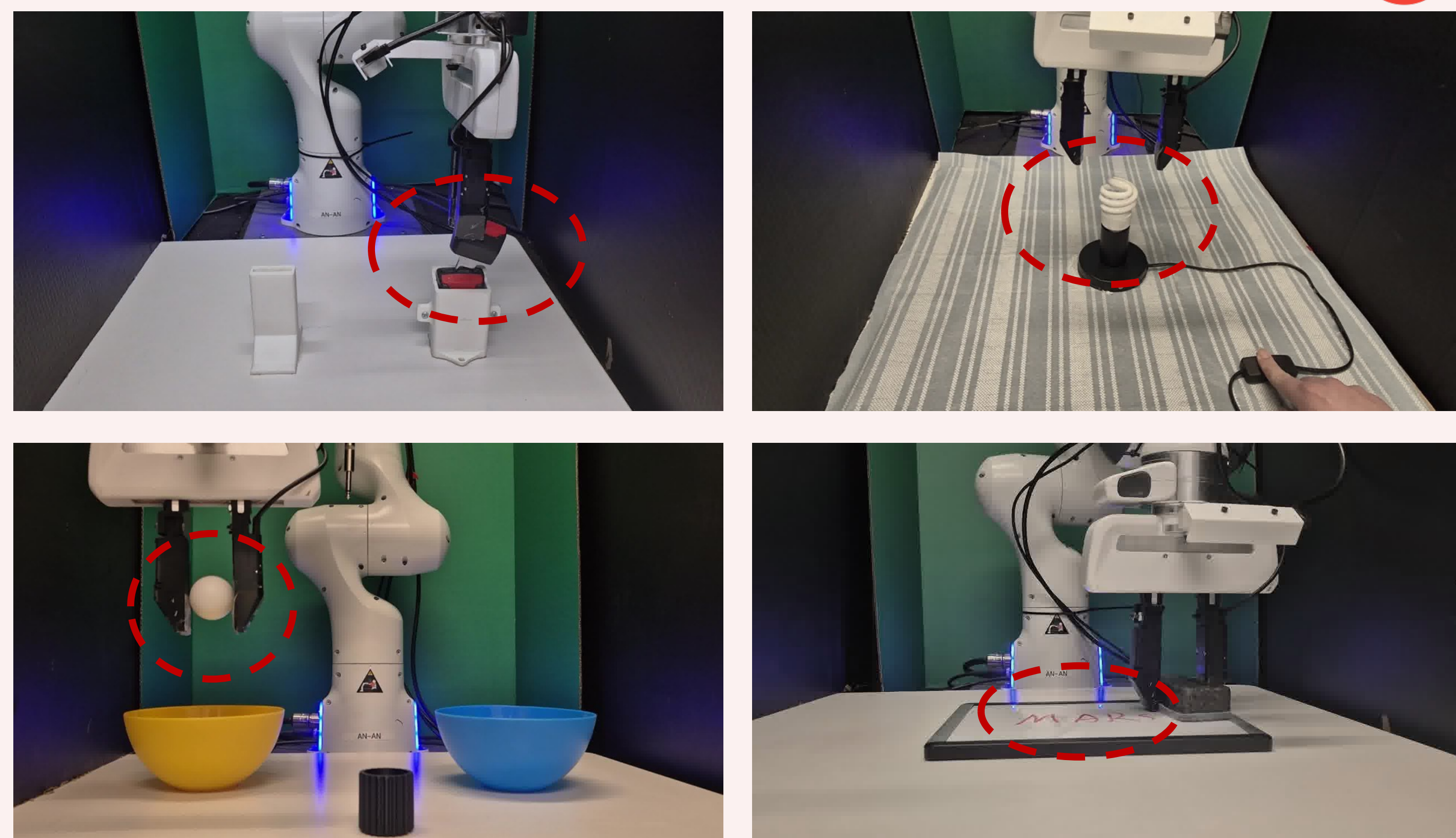
Can robots benefit from tactile knowledge without tactile sensors?

Vision+Tactile (with tactile sensors) ✓



High Performance

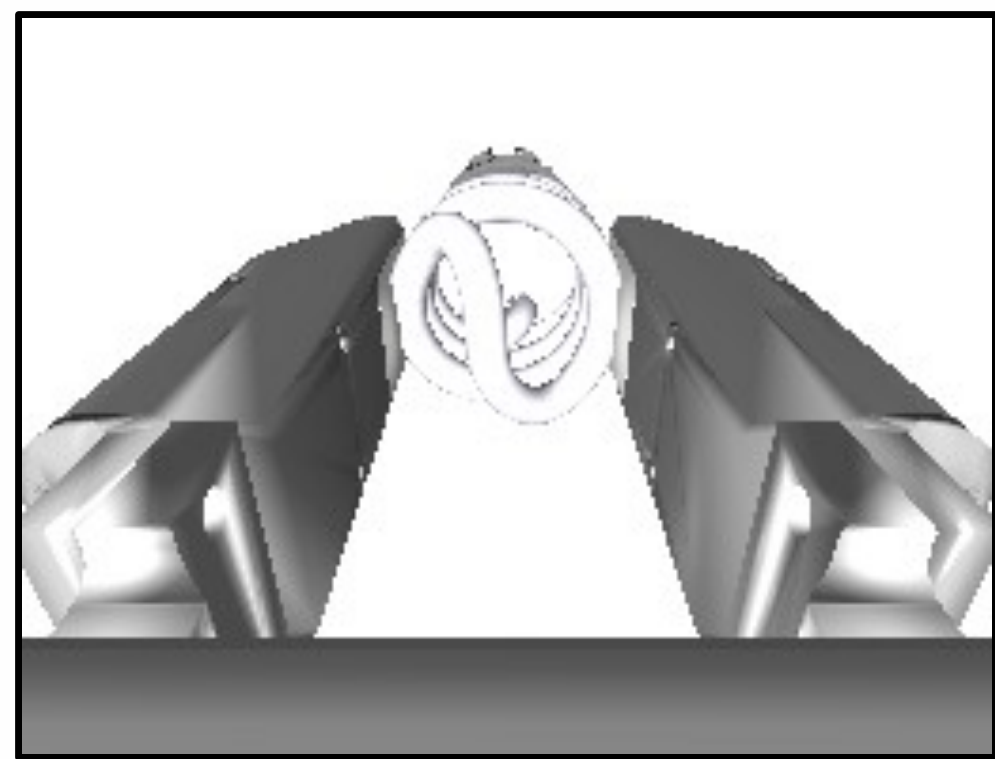
Vision Only (no tactile sensors) ✗



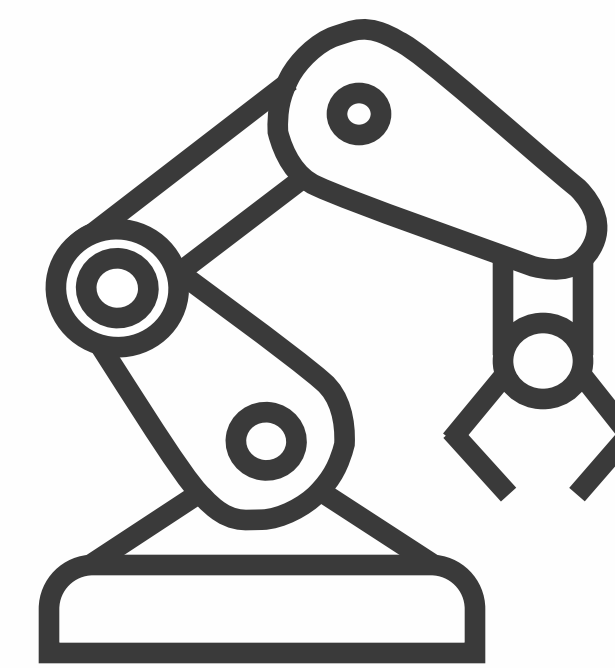
Contact Ambiguity → Performance Drop

(b) TacImag Framework

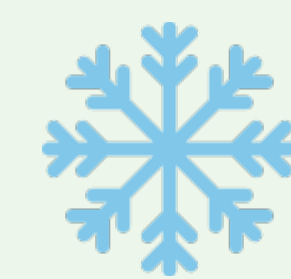
Vision



Proprioception

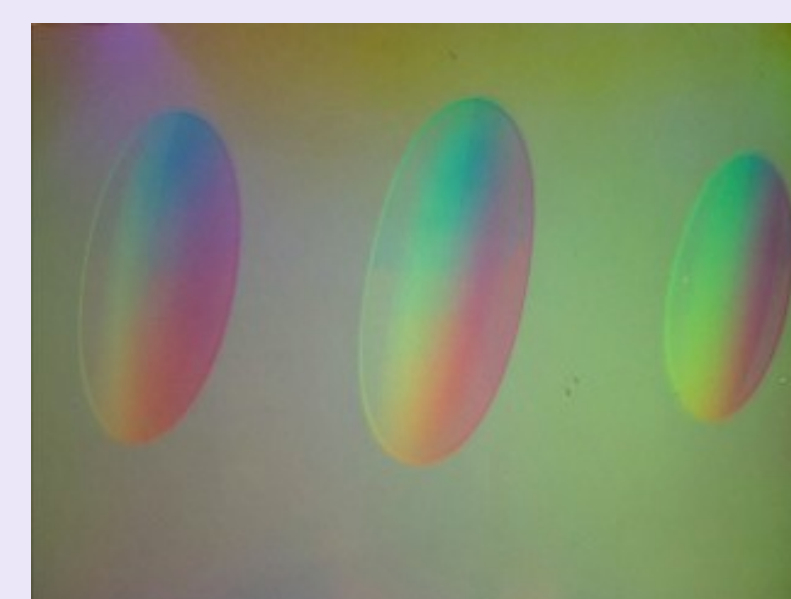


TacImag

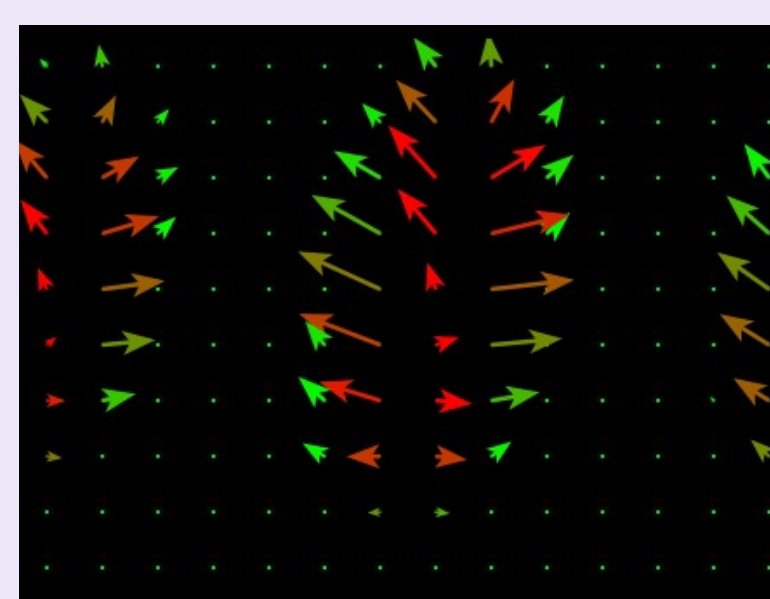


Tactile Imagination Model
(Frozen Diffusion Model)

Imagined Tactile Signal

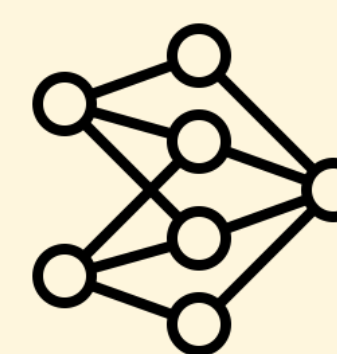


TacRGB



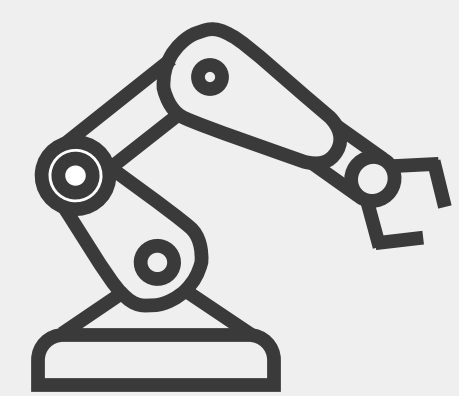
TacFF

Tactile-Conditioned
Manipulation Policy



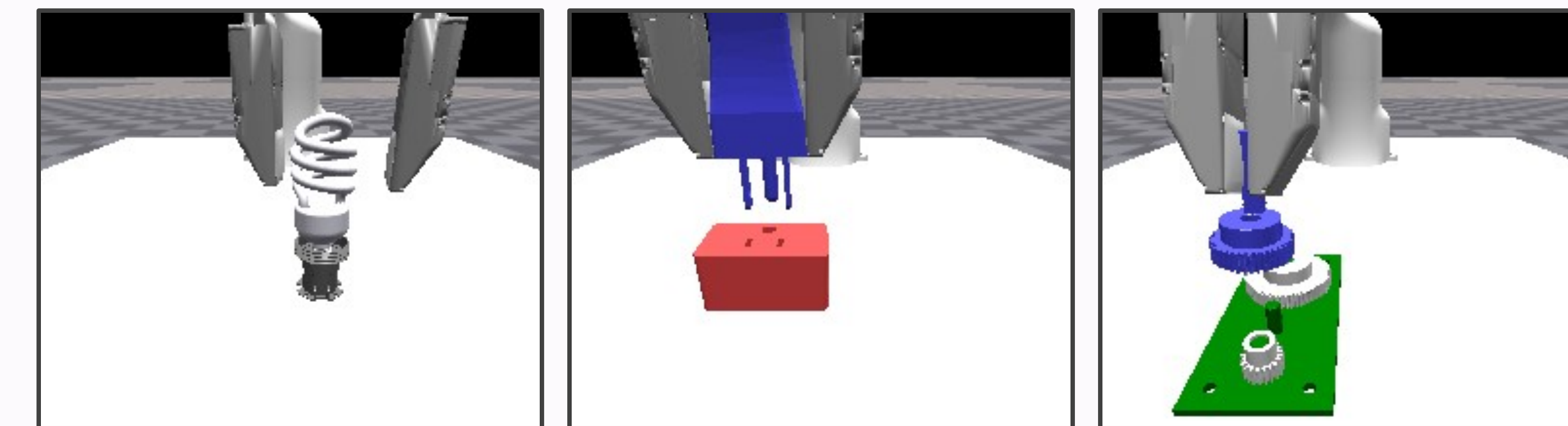
Action

(End-Effector Motion + Gripper)

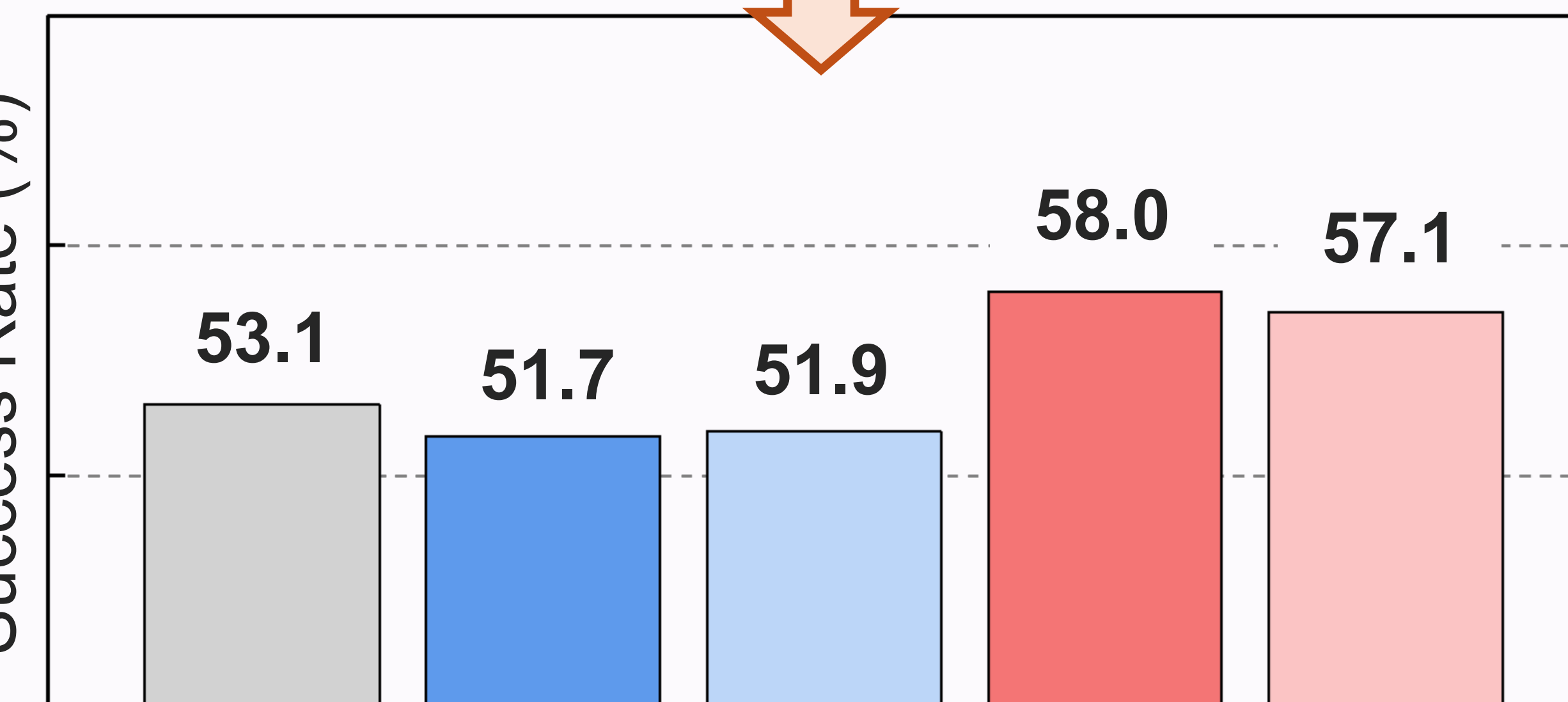


(c) Simulation Findings

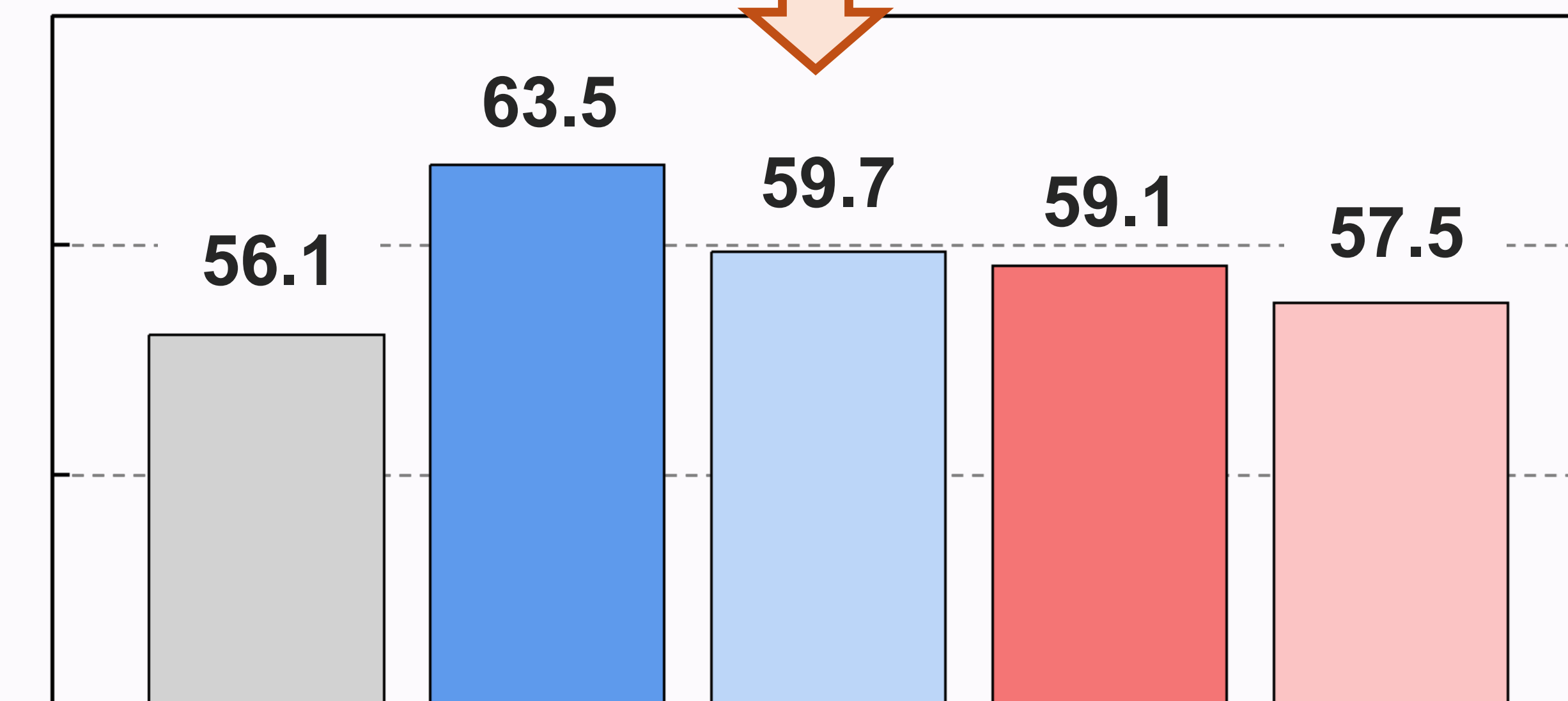
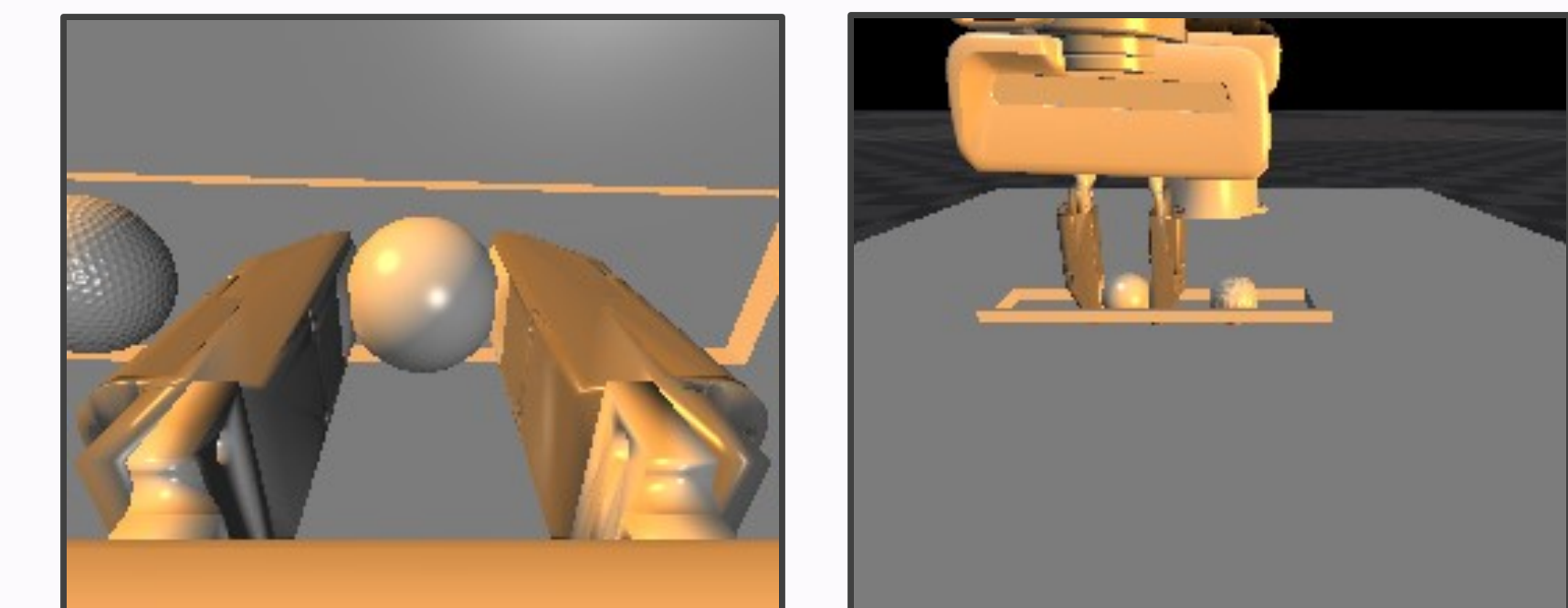
Contact-Sensitive Tasks



Success Rate (%)

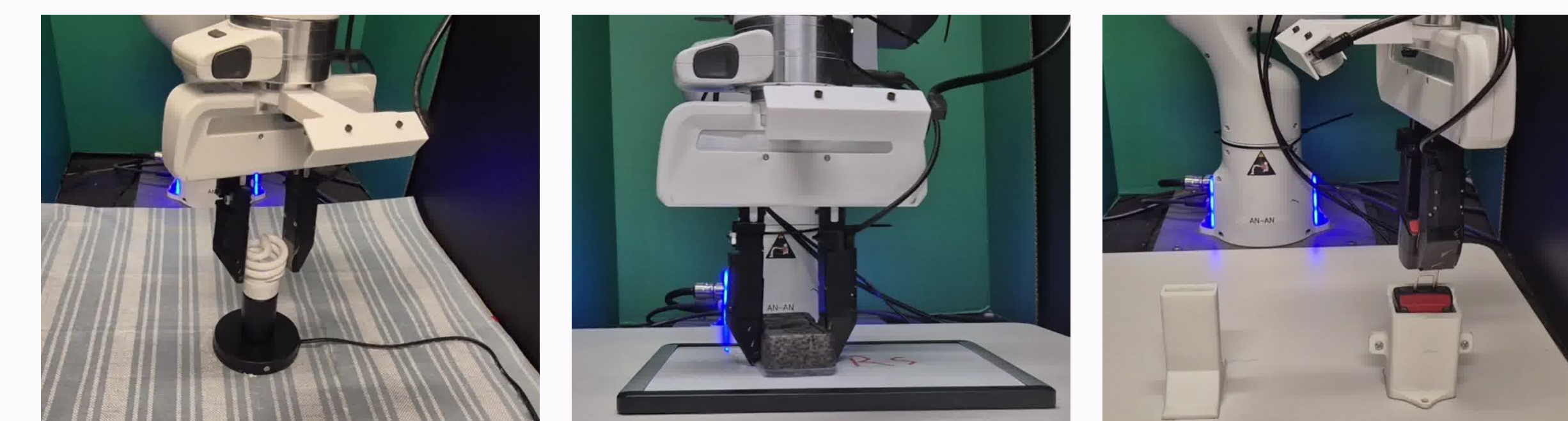


Texture-Sensitive Tasks

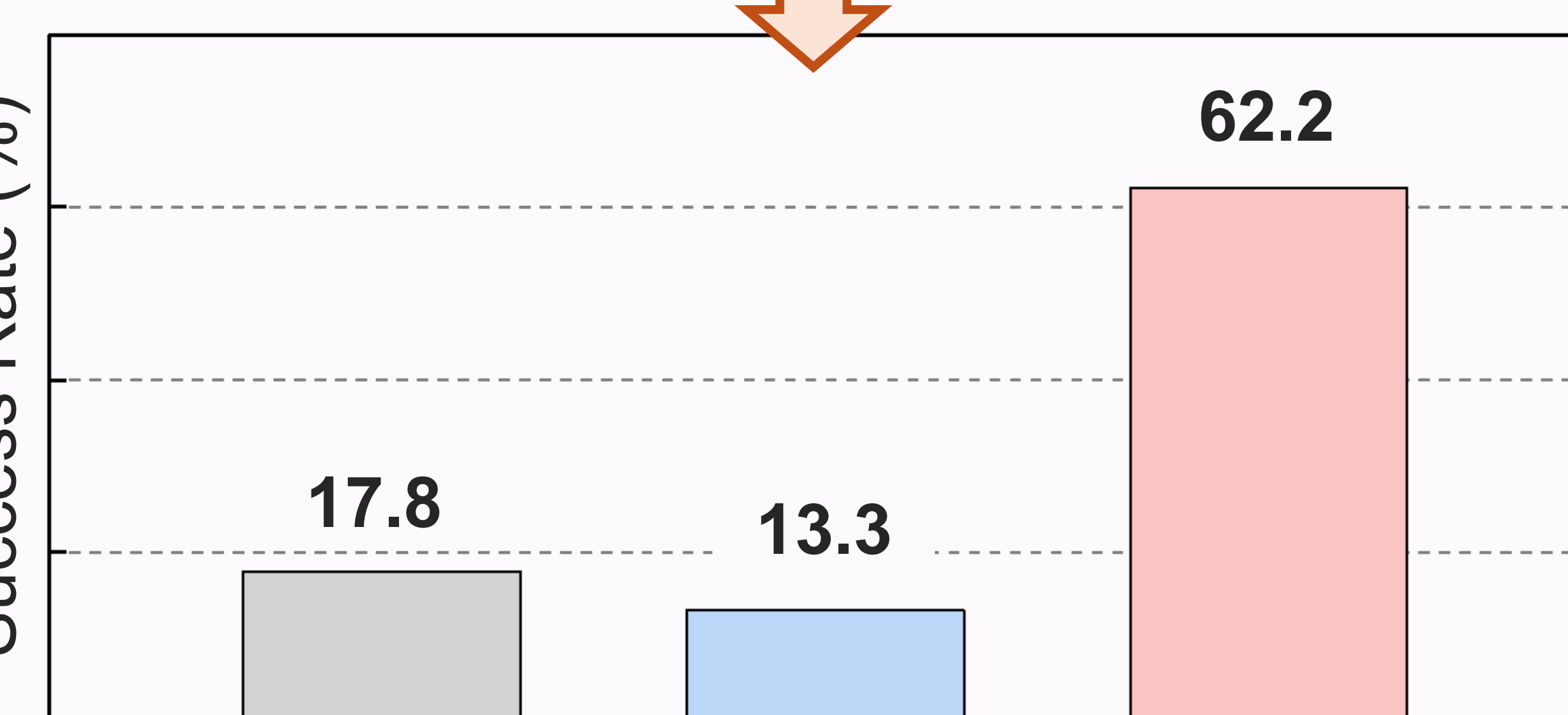


(d) Real-World Findings

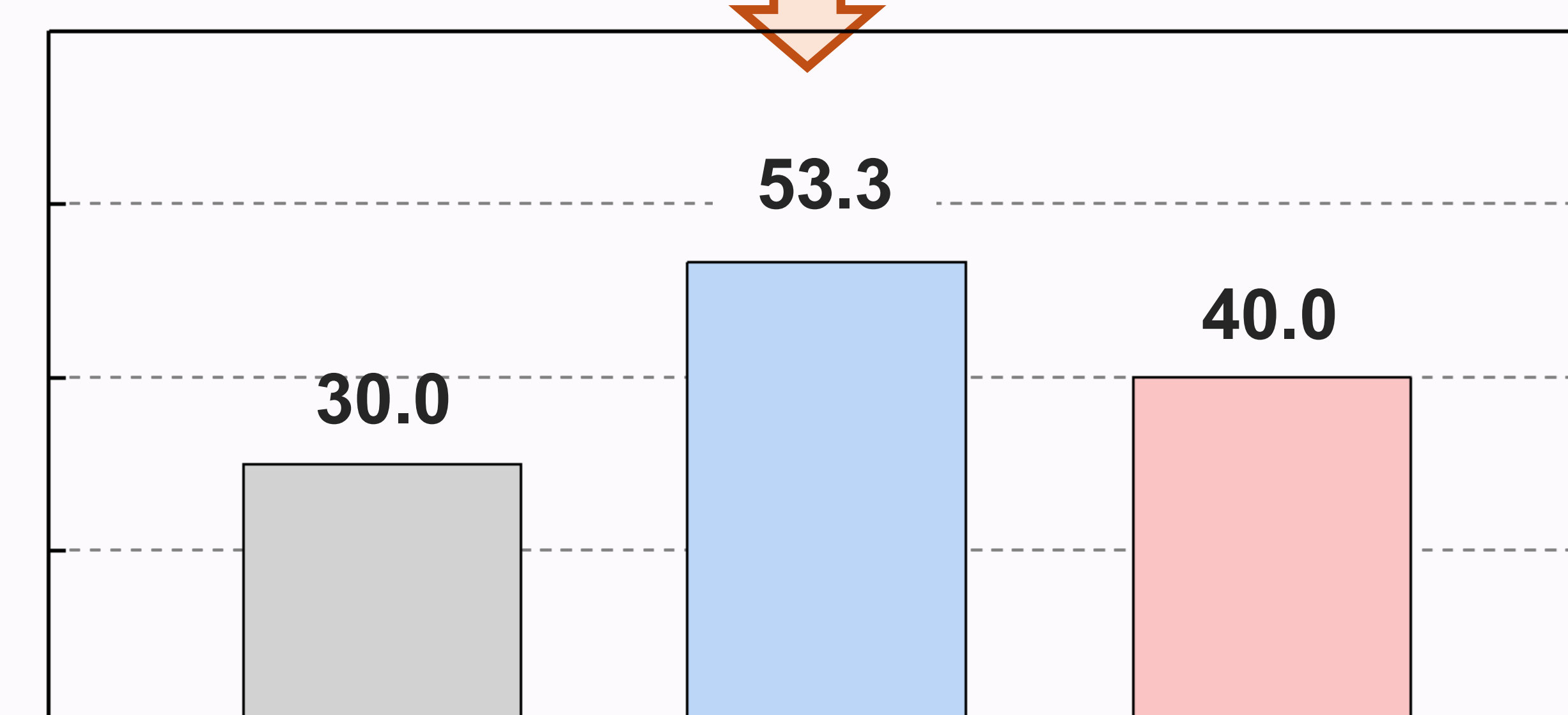
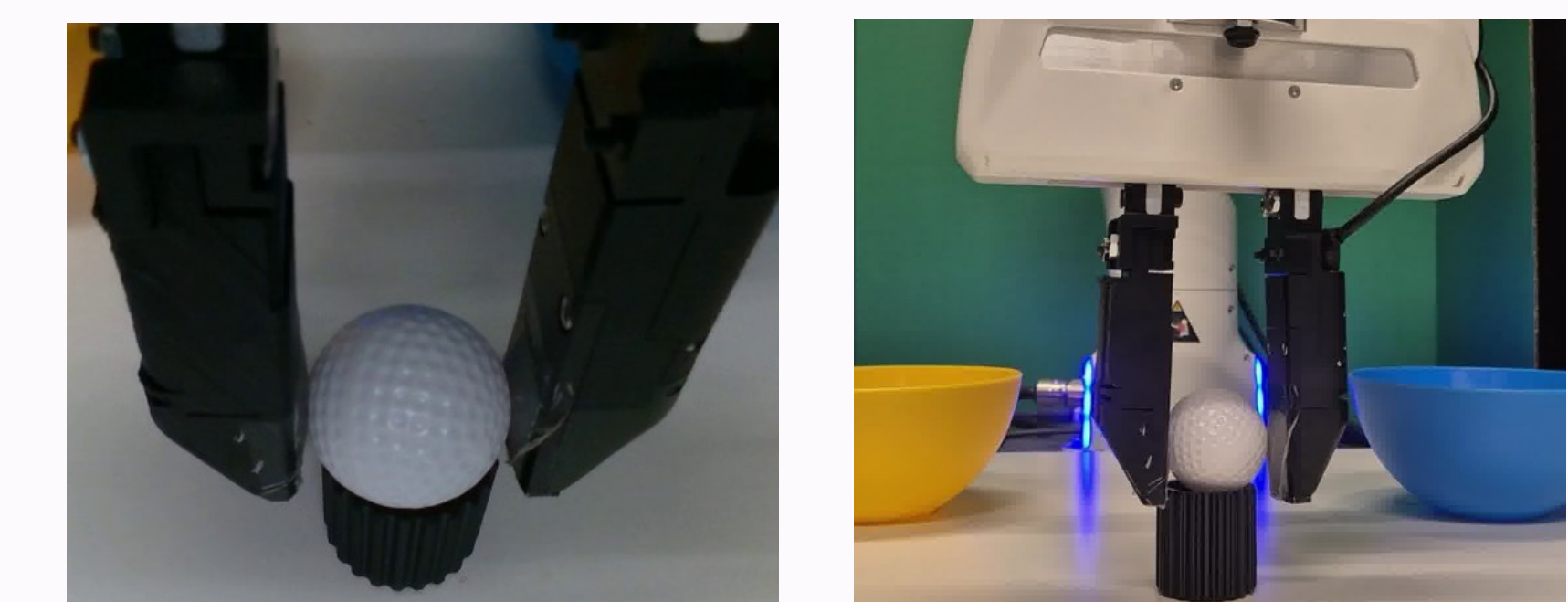
Contact-Sensitive Tasks



Success Rate (%)



Texture-Sensitive Tasks



Legend: Vision (grey), +TacRGB (Real) (blue), +TacRGB (Imag.) (light blue), +TacFF (Real) (red), +TacFF (Imag.) (pink)